Syed Tashfeenul Haque

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SUMMARY

ECE undergraduate with a solid foundation in **C++ and Python**, a strong inclination towards mathematics and analytical thinking. Actively studying **LLM and transformer architectures** — building models from scratch and applying concepts across real-world projects with a structured problem-solving mindset.

PROJECTS

Smart Solar Grass Cutter | Arduino ATmega328p, HC-SR04, IR Sensors, Bluetooth (HC-05), Proteus — Sep 2025 - Oct 2025

- Designed and built an autonomous solar-powered grass cutter integrating a **20W solar panel** with an ATmega328p microcontroller; achieved **6+ hours** of uninterrupted operation per charge cycle with zero external power dependency
- Implemented real-time obstacle detection using ultrasonic and IR sensors with a **30 cm detection threshold**, cutting false-trigger rate to under **5%** in field tests
- Enabled wireless Android app control over Bluetooth with an effective range of $\sim$**10 m**, reducing direct manual intervention by **80%** compared to conventional mowers
- Simulated and validated complete circuit in **Proteus**; recognized as the **best project in the batch** and highest-budget build at 6,000

Agentic AI Loan Assistant | FastAPI, Streamlit, Google GenAI (Gemini), Python, asyncio | GitHub — EY Techathon

- Architected a **5-agent orchestration pipeline** (Conversation $\rightarrow$ KYC $\rightarrow$ Credit $\rightarrow$ Offer $\rightarrow$ Document) with a FastAPI backend and Streamlit frontend themed for Tata Capital
- Integrated Google GenAI (Gemini 2.5 Flash) via an `asyncio.to_thread` wrapper, achieving average agent response latency of $\sim$**2–3 s**; reduced simulated loan intake cycle from $\sim$**25 min to under 9 min** end-to-end
- Implemented in-memory session persistence (optional Redis) and automated **PDF sanction letter generation**, cutting document turnaround from hours to $<$**10 s**
- Built a real-time **Agent Logs panel** in the Streamlit UI with timestamps and color-coded badges for live orchestration monitoring across all 5 agents

myChotuLLM | Python, PyTorch, Transformer Architecture — In Progress

- Building a **decoder-only transformer** from scratch in PyTorch — implementing scaled dot-product multi-head self-attention, learned positional encodings, and autoregressive text generation with greedy and top-k sampling
- Experimenting with model scales from **1M to 15M parameters**; targeting a character-level perplexity of $<$**45** on a curated text corpus with $\sim$10M tokens
- Profiling training throughput and memory utilization to optimize **batch size vs. sequence length** trade-offs, achieving $\sim$**3**$\times$ training speed-up with gradient accumulation on a single GPU

EDUCATION

Abdul Kalam Technical University — Aug. 2024 – May 2028 (Expected)

B.Tech – Electronics and Communication Engineering — New Delhi

- **Relevant Coursework**: Data Structures & Algorithms, Digital Electronics, Signals & Systems, Embedded Systems, Engineering Mathematics, Microcontrollers

SKILLS

- **Languages**: Python, C++, JavaScript, SQL
- **ML & AI**: PyTorch, Transformers, LLM Architecture, Google GenAI (Gemini)
- **Backend & Frameworks**: Node.js, Express, FastAPI
- **Frontend**: React, HTML, CSS, TailwindCSS
- **Databases**: MongoDB, SQL
- **Tools**: Git, GitHub, Postman, Proteus